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November 23, 1971
SUPERSEDING
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December 14, 1966

FEDERAL SPECIFICATION

TISSUE, DRY MOUNTING, PHOTOGRAPHIC

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specifcation covers the requirements for dry-mounting tissue for photographic use.

1.2 Classification. The dry-mounting tissue shall be of the following types:

Type I - Permanent.
Type II - Semi-permanent.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-636 - Boxes, Fiberboard.
PPP-B-676 - Boxes, Setup.
PPP-T-495 - Tubes, Mailing and Filing.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specification, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

FSC 6750

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

MILITARY SPECIFICATIONS:

MIL-P-116 - Preservation, Method of.

MILITARY STANDARDS:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 - Marking for Shipment.

(Copies of Military Specifications and Standards required by suppliers in connection with specification procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

American National Standards Institute (ANSI) Standards:

PH 1.12 - Photographic Sheet Paper for General Use.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, N.Y. 10018.)

National Motor Freight Traffic Association, Incorporated, Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 202, Union Station, 516 W. Jackson Blvd., Chicago, IL 60606.)

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3. REQUIREMENTS

3.1 Description. The dry-mounting tissue shall consist of a thin paper base uniformly impregnated, or coated on both sides, with an inert, flexible, thermoplastic adhesive. The tissue shall be dry to the touch, water resistant and shall not become soft or tacky at ordinary room temperatures, brittle when cold, nor crack or crumble.

3.2 Environmental storage. The tissue shall not be damaged by environmental heat or cold during shipment or storage within the temperature range of -50°F. to +150°F. Adhesive surfaces that tend to adhere to each other or to enterleaving paper during storage shall be easily separated without damage to th dry-mounting tissue.

3.3 Dimensions.

3.3.1 Sheets. Dimensions of cut sheets shall be in accordance with table I. Tolerances shall be +0, -1/32 inch for sizes up to and including 8 inches by 10 inches and +0, -1/16 inch for sizes greater than 8 inches by 10 inches. Squareness shall conform to the requirement of ANSI PH 1.12.

3.3.2 Rolls. Dimensions of roll tissue shall be in accordance with table I. Slitting width tolerances shall be +0, -1/16 inch. Tolerances for lengths up to and including 30 feet shall be +6 inches, -0 inch and tolerances for lengths of 100 feet and over shall be +18 inches, -0 inch.

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TABLE I. DIMENSIONS

Type I Nominal 0.003 Inch Thick				Type II Nominal 0.005 Inch Thick			
Sheets		Rolls		Sheets		Rolls	
Width (Inches)	Length (Inches)	Width (Inches)	Length (Feet)	Width (Inches)	Length (Inches)	Width (Inches)	Length (Feet)
4	5	20	30	4	5	16	100
5	7	20	150	5	7	32	100
8	10	40	30	8	10		
10	12	40	150	10	12		
11	14			11	14		
14	17			16	20		
16	20						

3.4 Performance. When placed between a photographic print and mounting board with sufficient heat and pressure applied, the adhesive shall flow into the paper base of the photograph and the mounting board fiber to form a dry bond. The tissue shall adhere firmly to the photo and mount when cooled to room temperature shall retain its original flexibility and shall not strip nor peel from the mount. The tissue shall not bleed nor impart noticeable color to either black and white or color prints or the mount.

3.4.1 Bonding.

3.4.1.1 Type I. Type I tissue shall form a permanent bond between a conventional black and white print and the mount within a temperature range of +200°F. to +275°F. after a dwell time in the press of not less than ten, nor more than twenty seconds. Color prints, and black and white prints on resin coated paper stock shall be permanently bonded to the mount within the temperature range of 210°F. to 230°F. after a dwell time of not less than 30, nor more than 60 seconds.

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3.4.1.2 Type II. When used with applied temperatures from 175°F. to 220°F. after a dwell time in the press of not less than 20 seconds, nor more than 40 seconds, type II tissue shall form a semi-permanent bond between the photograph and mount. When reheated to fusion temperature, the photograph shall be readily removable from the mount.

3.5 Workmanship. Workmanship shall be in accordance with good commercial practice in the manufacture of thermoplastic adhesives.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may use his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Lots. A lot for examination and tests shall consist of all the dry-mounting tissue of one type submitted for delivery at one time.

4.3 Sampling. Sampling for inspection and acceptance shall conform to the applicable provisions of MIL-STD-105.

4.3.1 Inspection levels. Sample sizes shall be selected according to inspection level S3 at an acceptable quality level of 2.5. The sample unit shall be one sheet of tissue for cut sheets and for roll tissue the sample shall be a square sample, each of the four sides of which is equal to the width of the roll.

4.4 Test Methods.

4.4.1 Physical examination. Inspect the tissue visually and with gauges to determine compliance with the requirements of this specification with respect to dimensions, squareness, uniformity of coating, and tackiness.

4.4.2 Water resistance. Immerse a specimen of the tissue in water at approximately 100°F. for 5 minutes. Remove the tissue, blot, air dry and use the specimen to mount a single weight print. Examine the mounted print for evidence of poor bonding or blisters.

4.4.3 Performance.

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4.4.3.1 Type I tissue. With a dry-mounting press heated to the lowest temperature specified herein for mounting black and white prints with type I tissue, mount a single weight and a double weight print with sample tissues. Increase the temperature of the press to the upper limit specified for black and white prints and mount a single weight and a double weight print. Verify that each of the prints has been properly bonded to the mount within the specified dwell period in the press. Repeat the test procedure with color prints on a resin coated paper base, using the high and low temperatures specified for resin coated paper, and verify that a proper bond can be attained within the specified dwell period. Flex each of the mounted prints approximately 30° along each diagonal, first with the print on the inside, and then with the print on the outside of the radius of curvature and verify that the print remains bonded to the mount. Loosen one corner of each print and forcibly pull the print from the mount. Verify that paper fibers from the print or the mount adhere to the tissue.

4.4.3.2 Type II tissue. Using type II tissue and the temperatures specified for type II, repeat the test procedures specified for type I tissue to mount conventional black and white prints as well as color prints on a resin coated paper base. Verify that a proper bond can be attained within the specified dwell period. Flex each of the mounted prints approximately 30° along each diagonal, first with the print on the inside, and then with the print on the outside of the radius of curvature and verify that the print remains bonded to the mount. Reheat the print to approximately 200°F. and immediately peel the print from the mount. Verify that no paper fibers adhere to the tissue.

4.4.4 Environmental Storage. Stack the contents of one package of tissue for a period of forty-eight hours at a temperature of minus 50°F., +5°F., under a suitable weight to maintain a uniformly distributed pressure of approximately 0.1 pound per square inch over the surface of the tissue to increase the blocking tendency. Interleaving papers may be used provided they are included in the manufacturer's commercial packaging practice. At the end of the cold storage period, raise the temperature of the storage environment to 150°F., for a period of forty-eight hours. Remove the tissue from the test chamber and allow the stack to stabilize at a room temperature of approximately 70°F. Upon completion of the test period, verify that the sheets of tissue can be easily separated from each other, or from the interleaving paper without damaging the tissue. Using sample sheets selected from those subjected to the storage environment, repeat the applicable performance tests specified herein (see 4.4.3.1 and 4.4.3.2), and determine that the tissue meets the specified performance requirements.

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4.5 Inspection of Preparation for Delivery. An examination shall be made to determine that markings, materials, workmanship, contents, and weights comply with the requirements of section 5 of this specification. The sample unit shall be one shipping container, fully packed, selected just prior to the closing operation. The inspection level shall be S-2 and the acceptable quality level 4.0 defects per hundred units in accordance with the applicable provisions of MIL-STD-105.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C,as specified (see 6.2).

5.1.1 Level A. Type I sheets in quantities of 150 or 500 (see 6.2). or type II sheets in quantities of 145 four sheets, shall be packaged in a box conforming to PPP-B 676, type I, variety 1, class A. Style R. Roll tissue shall be wound on a core of the type normally used for the product and packaged in a close-fitting box conforming to PPP B 636, class domestic or in a tube conforming to PPP-T-495. The packages shall be secured to prevent accidental opening.

5.1.2 Level C. The tissue shall be packaged in accordance with the supplier's commercial practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The tissue, packaged as specified in 5.1.1, in quantities as specified (see 6.2), shall be packed in a close fitting box conforming to PPP-B-636, class weather-resistant. The box shall be closed and water-proofed in accordance with the appendix to the box specification.

5.2.2 Level B. Unit packages in quantities as specified (see 6.2), shall be packed in a close-fitting box conforming to PPP-B-636, class domestic. The box shall be closed in accordance with the appendix to the box specification.

5.2.3 Level C. The tissue, packaged as specified in 5.1 shall be packed in containers to assure carrier acceptance and safe arrival at destination in compliance with Uniform Freight Classification or National Motor Freight Classification.

5.3 Marking. In addition to any special marking required in the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No 123 for civilian agencies or MIL-STD-129 for military agencies. Markings shall include the statement: "Store in a Cool, Dry Place."

6. NOTES

6.1 Intended Use. The dry-mounting tissue covered by this specification is intended for use in mounting photographic prints onto cardboard mounts.

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6.2 Ordering Data. Purchasers should exercise the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.

(b) Type of tissue desired (see 1.2, table I).

(c) Whether roll or cut sheets (see 3.3.1, 3.3.2, table I).

(d) Nominal size of sheets or rolls (see 3.3.1, 3.3.2, table I).

(e) Quantity of sheets per unit package (see 5.1.1).

(f) Quantity of unit packages required (see 5.2.1 and 5.2.2).

(g) Levels of packaging and packing (see Section 5).

MILITARY INTEREST:

Custodians:
Army - EL
Navy - AS
Air Force - 70

Review Activities:
Army - EL
Navy
Air Force -

User Activities:
Army - EL
Navy
Air Force

Preparing Activity:

Navy - AS

CIVIL AGENCY INTEREST:

GSA- FSS

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